

# Analyst-Centered Evaluation: A User Perspective on the Effectiveness of Provenance Graphs in Alert Investigation

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## Abstract

Provenance-based intrusion detection systems (PIDS) have shown great promise in detecting Advanced Persistent Threat (APT) activities in lab setting. Since they capture system behavior as a directed graph, the alerts generated by such PIDSs also naturally show causal relationships between the system entities (processes, files, and network connections). Provenance graphs are therefore assumed to make alert investigation more interpretable for security analysts, in which case, they can prove as an effective tool in the investigation workflow. However, we see a disconnect: 1) PIDS works mainly evaluate their systems without the human analyst in the loop who must take action on the generated alerts and investigate them. 2) Tabular view of row-column structure is the default choice in alert investigations for the same human analysts. We do not know how useful provenance graphs are for this task. This leaves an open question: *To what extent do provenance graphs aid security operators in alert investigations?*

We address this gap with a within-subjects, counterbalanced user study (N=16) in which participants perform a task we call attack artifact discovery. Participants investigate attack scenarios using two approaches: once using provenance graphs and once searching through traditional logs in table. Given the attack scenario, the participant has to trace the attack steps and identify the system entities and events involved (e.g. processes spawned, files touched, connections opened). We define the success of attack investigation in terms of *quality* and *efficiency*. Our results show that participants using the provenance graph performed significantly better in efficiency and collected 2.8X more attack artifacts than using tables. Provenance graphs did not improve the quality of the collected attack artifacts significantly but it needed less effort to collect attack artifacts with, cutting median time per correctly identified artifact by 55%. Finally we share our design decisions and practical insights which might be interesting for the industry and future researchers interested in adopting provenance graphs for security alert investigations.

## CCS Concepts

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## 1 Introduction

On a system/host level, provenance data helps in keeping detailed record of the events that take place [28] and hence is crucial evidence for root-cause analysis [29]. Provenance data is captured as a sequence of triplets in the form *subject-operation-object*. Each triplet captures which entity acted (subject), what actions was performed (operation) and on which entity (objects). For example in figure 1, the triplet (*usr/bin/firefox*, *clone*, *usr/bin/firefox*) means that *usr/bin/firefox* process clones to a new process. Subsequently, this process reads the file */etc/passwd* (*usr/bin/firefox*, *read*, */etc/passwd*) and then sends information to a suspicious IP address (*usr/bin/firefox*, *sendto*, *146.153.68.151:80*).

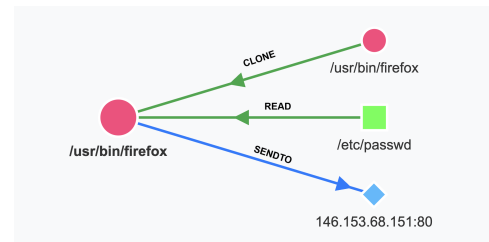


Figure 1: Example of provenance graph in our ACE platform.

The triplet of (*subject*, *operation*, *object*) in system provenance data can also be organized into a directed graph called provenance graph [39]. In the provenance graphs, nodes represent entities like files, processes and network connections with the edges representing interactions or operations between such entities. This encodes causality, e.g. in figure 1 following an edge from the suspicious network connection backwards (backward analysis) can reveal the process that initiated it, which in our case is */usr/bin/firefox*. Similarly, following this process forward (forward analysis) reveals what that process read (*/etc/passwd*) or executed.

This is the basis of PIDSs which convert system provenance data into a provenance graph and apply anomaly detection [4, 17] or rule-based detection [11, 14, 34] to detect attacks. The alerts generated by such PIDSs are structured as subgraphs of entities and operations which form the suspicious activity. Compared to a flat list of log lines detected in a traditional IDS, a provenance graph of alert provide security operators more context about the event or the chain of events that make up the suspicious activity on a system [15].

The implicit claim is that presenting provenance data as a graph, rather than tabular event logs enables security operators to understand the attack context more readily, and investigate alerts more effectively. This is a reasonable hypothesis because graph structures indeed capture the causal relationships a security operator need to follow. Since provenance graphs connect causally-related alarms, they potentially reduce the manual effort of connecting the dots across log sources.

However, this has not been validated empirically. The assumption that graph structures aids alert investigation is foundational to the design of PIDSs output format. Moreover, existing PIDS evaluations focus on detection performance against a ground truth in commonly used datasets such as from DARPA Transparent Computing (TC) program [5, 6]. Whether a security operator can effectively use a provenance-graph to investigate an alert— which includes finding the relevant artifacts, understanding the attack chain and making a correct escalation decision, remains an open question. Industry acknowledges that provenance-based intrusion detection systems are more interpretable [7]. Provenance graphs offers a rich and detailed causal representation between the connected system entities. On the contrary, tables do not capture the relationships between the system entities automatically. So if this structural advantage of provenance graphs is real, it should also appear as better performance in alert investigation itself.

During an alert investigation [22], the security operator has to find which events connect to which— this includes finding out what process initiated the suspicious activity, which files were accessed and what network connections were made. Such system events and entities are tangible indicators providing direct evidence of a threat activity. We refer to these indicators specifically as *attack artifacts* [30]. In comparison to Indicators of Compromise (IoC) where the goal is to share threat intelligence [21], attack artifacts are tied to the host and refers to local, session-specific system entities for e.g. child processes spawned by a process or local network sockets. In figure 1, the IP address can be an IoC as it is related to the attacker’s infrastructure. On the other hand, the process `/usr/bin/firefox` is an attack artifact but not an IoC. Essentially, all IoCs are attack artifacts but not all attack artifacts can be IoCs.

We hypothesize that provenance graph should aid security operators in a more *complete* and *timely* discovery of attack artifacts. Based on this criteria, we define two keys elements to measure success in alert investigation: *quality*— whether security operator finds attack-relevant artifacts and hence performs a complete investigation; *efficiency*— whether security operator finds these right artifacts timely. In essence, we ask the questions:

**RQ1:** To what extent do provenance graphs affect the *quality* of attack artifacts collected in alert investigation?

**RQ2:** To what extent do provenance graphs affect the *efficiency* of a security operator in alert investigation?

To answer these questions, we conduct a within-subjects study in which participants performed alert investigation using two interfaces: an interactive provenance graph and an equivalent searchable tabular log view. We also built ACE (Analyst-Centered Evaluation) to support this study. ACE is a platform that exposes the same provenance data for an alert in these two interfaces. Using ACE, participants investigated alerts and collected attack artifacts which are part of the suspicious activity. Results show that participants

found significantly more attack artifacts in the same amount of time with provenance graphs. However, the same participants did not see any significant improvement with respect to the quality of the collected attack artifacts. Surprisingly, participants did not need any special training for working with provenance graphs. This suggests that provenance graphs can be incorporated to existing tool-set of security operators without long training times. Our findings also support the claim about interpretability of provenance graphs [7] but goes one step further to validate it and see what practitioners stand to gain from them in the task of alert investigation.

We thus make the following contributions:

- **A controlled user study with task-based evaluation of provenance-graph for alert investigation.** We formulate alert investigation as an attack-artifact discovery task and conduct a within-subjects study with 16 participants comparing their performance on provenance graph against an equivalent tabular view.
- **Empirical evidence that choice of interface affects efficiency more than accuracy.** Participants using provenance graphs completed more of the investigation, collected more artifacts and spent less time doing so. Quality metrics (precision, recall, F1) showed no statistically significant difference between interfaces.
- **ACE, an open-source platform** with interactive provenance graphs and equivalent table view over identical provenance data, to enable future human-centered studies.

In summary, we examine if the better interpretability of provenance graphs results in better performance in alert investigation. Results indicate that although provenance graphs can help security operators move fast in an alert investigation and collect more attack artifacts, it does not necessarily improve the quality of the collected attack artifacts.

We believe our study design can motivate future works and we will make the code and analysis scripts open-source. The rest of the paper is structured as follow. We discuss the experiment setup and task in section 2. Then we analyze the study results in section 3 and discuss the implications along with some practical insights in section 4. Finally we take a look at related work in 5 and conclude in section 6.

## 2 Methodology

We focus on the investigation stage that follows when an alert is raised. While a false positive is documented and closed, a suspicious alert has to be investigated further. For such alerts, a security operator collects the related attack artifacts and reconstructs the events that fired the alert for root cause analysis. Investigation therefore broadly involves finding artifacts and connecting them into a chain of events.

We judge performance on such investigation on two axes: whether the operator collects correct artifacts, and whether they collect enough of them in time. RQ1 asks whether the interface affects the quality of artifact discovery, measured through precision, recall, and F1-score. RQ2 asks whether the interface affects efficiency – the coverage and speed with which discovery takes place, measured

through four metrics explained in Section 2.3. Both questions concern the same investigation stage: an analyst reconstructing what happened and assembling the evidence that justifies further action.

The goal is to compare participant performance on attack artifact discovery when using table versus provenance-graph interfaces. We employ a within-subjects design where [10], rather than randomly assigning each participant to strictly one condition (table or provenance graph), every participant experiences both (table and provenance graph).

## 2.1 Task: Attack Artifact Discovery

We define *attack artifacts* as the comprehensive set of concrete, verifiable digital footprints left behind by threat activity within a compromised infrastructure. We distinguish between two categories: general Indicators of Compromise (IoCs) and system-specific attack artifacts. IoCs are standardized, threat-intelligence-aligned signatures such as known malicious IP addresses or file hashes. On the other hand, attack artifacts also include broader, volatile host-specific data such as active process IDs (PIDs), memory dumps, or local network sockets involved in the attack. IoCs are therefore, a subset of the total attack artifacts left on a system.

For example, a Process ID (PID) is an attack artifact because it objectively proves that a program ran as part of threat (or benign) activity. It is not an IoC, because PIDs are randomized by the operating system and cannot be shared as static threat intelligence to protect other networks.

We chose attack artifact discovery as our task because it matches what an operator already does during investigation, and because it can be scored objectively. An operator collecting evidence for triage or escalation is, at each step, deciding whether something in front of them is or isn't part of the attack. That decision is binary and verifiable against a fixed attack scenario. Hence we can build ground truth of attack artifacts for a chosen attack scenario and score participants against that ground truth directly, without relying on self-report or subjective judgment of "how well" someone investigated.

**Attack scenario.** Our methodology assumes host-level system provenance data: a record of process, file, and network events on a compromised host. We use the DARPA TC Engagement-3 dataset [5] of THEIA host as the attack scenario. It is a widely-used benchmark in PIDS works and provides validated ground truth of the attack artifacts. Two alerts were selected from this dataset, each flagged on a netflow node representing suspicious communication with an external IP address.

**Assessment questions.** The questions follow the idea of forward analysis where the investigation start with an entity as a starting point. For each alert, this starting point is a network connection which represent the initial communication between the host and an attacker's IP address. From there, participants answer nine structured questions per alert, using the assigned interface to collect evidence and reconstruct the attack as it unfolds.

Because we know the ground-truth artifacts for each question in advance, we can score every response directly: artifacts the participant reports that are part of the attack count as true positives (TP), artifacts they miss count as false negatives (FN), and artifacts they report that are not part of the attack count as false positives (FP).

For instance, table 1 shows the performance of a participant in each question. In questions Q1 and Q3, the participant successfully identified all the attack artifacts relevant to that stage in investigation. In question Q5 however, the participant missed 10 attack artifacts and reported a total of four attack artifacts where only two were part of the attack. This scoring gives us two things. First, it lets us derive traditional metrics like precision, recall and F1-score without relying on self-reporting. Second, resulting scores are directly comparable across participants and between the two interfaces of investigation— provenance-graph and table.

**Table 1: Per-question TP/FP/FN counts for a participant (provenance graph interface, Alert #1, Q1–Q5 and rest Q6–Q9 unattempted). The column *Submitted* shows the number of attack artifacts reported by the participant. The column *Ground Truth* contains the actual number of attack artifacts involved.**

| Question | TP | FP | FN | Submitted (TP+FP) | Ground Truth (TP+FN) |
|----------|----|----|----|-------------------|----------------------|
| A1-Q1    | 1  | 0  | 0  | 1                 | 1                    |
| A1-Q2    | 7  | 0  | 3  | 7                 | 10                   |
| A1-Q3    | 2  | 0  | 0  | 2                 | 2                    |
| A1-Q4    | 0  | 1  | 1  | 1                 | 1                    |
| A1-Q5    | 2  | 2  | 10 | 4                 | 12                   |

## 2.2 Hypotheses

We are motivated by one observation that provenance graphs encode the causal structure in events. In comparison, a security operator must reconstruct this causality manually from flat logs. Therefore, participants using provenance graphs should perform better in alert investigation itself. We formalize this as two hypotheses with regard to our devised task of attack artifact discovery.

**H1.** Provenance graphs increase quality in attack artifact discovery relative to tables, measured as precision, recall, and F1 over the collected artifacts.

**H1.** Provenance graphs increase efficiency relative to tables, measured as question coverage, artifact coverage, time per question, and time per artifact.

The corresponding null hypotheses are:

**H1<sub>0</sub>.** Performance between provenance graphs and tables do not differ in terms of quality in attack artifact discovery.

**H2<sub>0</sub>.** Performance between provenance graphs and tables do not differ in terms of efficiency in attack artifact discovery.

## 2.3 Metrics

We are interested in measuring performance in alert investigation under two categories: *quality*, how correct a participant's identified attack artifacts were (RQ1), and *efficiency*, how much ground they

covered and how quickly (RQ2). *Quality* in alert investigation cannot be captured by a single number. A participant who only reports artifacts they are certain about will look accurate but thorough investigation is not guaranteed. Conversely, a participant who reports everything plausible will look thorough but many of those reported attack artifacts will be wrong. Therefore, precision and recall each capture one side of this trade-off, and F1 combines them into a single score.

*Efficiency* has two parts: how much of the investigation was covered, and how much time the investigation took. Counting only the questions a participant attempted will not capture how much of the underlying attack was actually uncovered. Two participants can attempt the same number of questions and still recover different numbers of attack artifacts. We therefore split coverage into two metrics: *question coverage*, the number of questions attempted, and *artifact coverage*, the number of attack artifacts correctly identified relative to the total ground truth for that alert. Similarly, total time spent is an incomplete measure, since a participant who stops early after answering few questions can appear fast without having covered much ground. We therefore split time into *time per question*, elapsed time divided by the number of questions attempted, and *time per artifact*, elapsed time divided by the total number of ground-truth artifacts for that alert. Coverage and time together give four metrics: question coverage, artifact coverage, time per question, and time per artifact.

This gives us a total of seven metrics and these are computed per alert  $a$  and interface  $c$  (provenance-graph, table). Four of the total 18 questions across the two alerts have no defined artifact ground truth and are excluded from every metric below.  $Q_a$  denotes the remaining question set for alert  $a$ , with  $|Q_a| \leq 9$ . For each question  $q \in Q_a$ , a response is compared against a ground-truth artifact set, yielding counts of true positives ( $TP_q$ ), false positives ( $FP_q$ ), and false negatives ( $FN_q$ ). We further write  $Q_a^{\text{att}}(p, c) \subseteq Q_a$  for the subset of questions a participant  $p$  actually attempted under interface  $c$ . We now define each metric formally.

**2.3.1 Precision.** Precision measures the correctness of a participant's identified artifacts. It is computed only over attempted questions, so an unattempted question neither helps nor penalizes this metric:

$$\text{Precision}(a, c) = \frac{\sum_{q \in Q_a^{\text{att}}} TP_q}{\sum_{q \in Q_a^{\text{att}}} (TP_q + FP_q)}$$

A higher precision means more of a participant's reported artifacts were correct and attack-relevant.

**2.3.2 Recall.** Recall measures the thoroughness of a participant's investigation, over the same attempted questions:

$$\text{Recall}(a, c) = \frac{\sum_{q \in Q_a^{\text{att}}} TP_q}{\sum_{q \in Q_a^{\text{att}}} (TP_q + FN_q)}$$

A higher recall means a participant identified a larger share of the ground-truth artifacts relevant to the questions they attempted.

**2.3.3 F1.** F1 is the harmonic mean of Precision and Recall, and gives a single value:

$$F1(a, c) = \frac{2 \cdot \text{Precision}(a, c) \cdot \text{Recall}(a, c)}{\text{Precision}(a, c) + \text{Recall}(a, c)}$$

F1 penalizes an imbalance between precision and recall, so a high F1 requires a participant to be both correct and thorough on the questions attempted.

**2.3.4 Question Coverage.** Question coverage measures how much of the assessment a participant attempted, independent of correctness:

$$\text{QuestionCoverage}(a, c) = |Q_a^{\text{att}}(p, c)|$$

reported as a raw count out of  $|Q_a|$ . A higher question coverage means a participant engaged with more of the fixed question set for that alert.

**2.3.5 Artifact Coverage.** Artifact coverage measures the share of ground-truth artifacts a participant uncovered across the *entire* question set, attempted or not. An unattempted question is not excluded here, and is scored as a complete miss,  $TP_q = 0$  and  $FN_q$  equal to the full ground-truth set for  $q$ :

$$\text{ArtifactCoverage}(a, c) = \frac{\sum_{q \in Q_a} TP_q}{\sum_{q \in Q_a} (TP_q + FN_q)}$$

This is the check on Recall metric described earlier, so a participant who skips questions can have a high Recall (by only collecting attack artifacts that they are confident about) while their Artifact Coverage stays low, because most of the attack scenario went unaddressed.

**2.3.6 Time per Question.** Time per question measures pace in investigation over attempted questions. Let  $\text{Time}(a, c)$  be the time spent on  $Q_a^{\text{att}}(p, c)$ . Then:

$$\text{TimePerQuestion}(a, c) = \frac{\text{Time}(a, c)}{|Q_a^{\text{att}}(p, c)|}$$

A lower value means a participant moved through the questions they attempted more quickly.

**2.3.7 Time per Artifact.** Time per artifact measures how efficiently a participant found a correct attack artifact, using only successful finds. Let  $TP_{\text{found}}(a, c) = \sum_{q \in Q_a^{\text{att}}} TP_q$  be the total true positives across attempted questions. Then:

$$\text{TimePerArtifact}(a, c) = \frac{\text{Time}(a, c)}{TP_{\text{found}}(a, c)}$$

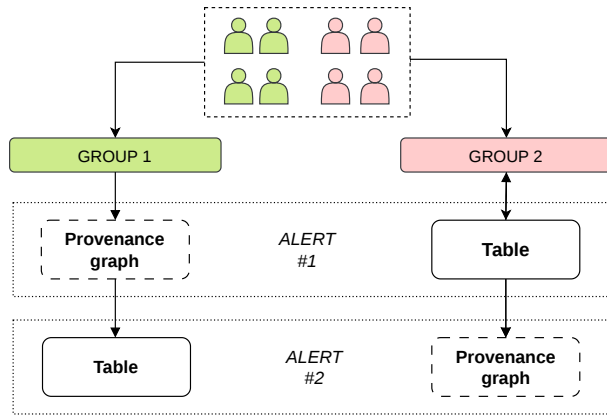
Using  $TP_{\text{found}}$  rather than a raw attempt count means time spent on wrong or empty answers does not make a participant look more efficient.

Precision, Recall, and F1 thus correspond to RQ1, the *quality* of attack artifact discovery, while Question Coverage, Artifact Coverage, Time per Question, and Time per Artifact correspond to RQ2, its *efficiency*.

## 2.4 Experiment Design

Figure 2 provides an overview of the experiment design. We employ a within-subjects design in the experiment. Every participant investigates one alert with the table interface and the other alert with the provenance-graph interface, rather than being assigned to just one condition.

*Experiment variables.* The independent variable is the interface: table or provenance graph. The dependent variables are the seven metrics defined in Section 2.3, computed separately for each interface a participant used.



**Figure 2: Overview of experiment design**

Every participant in each group completes the task of attack artifact discovery, using both the table and provenance-graph interface. Group 1 (provenance-graph  $\rightarrow$  table) and Group 2 (table  $\rightarrow$  provenance-graph). The sequence is switched between groups to counterbalance order. Separate alerts mitigate the learning effect.

**2.4.1 Participants.** Participants were recruited from a cybersecurity focused course at the university. They are graduate students with coursework in anomaly detection, attack graphs and provenance graphs. The study was done as part of a course assignment and participants were given access to the ACE tool the day before, to get familiar with the platform. Participation was voluntary and 21 students consented to be included in the study. Five participants were excluded because they did not follow the assigned procedure – one used the table interface for both alerts, while the rest mixed provenance graph and table within a single alert instead of keeping one interface per alert. We focus on the remaining 16 participants for our analysis.

**2.4.2 Procedure.** The purpose of the study was disclosed upfront to all participants. The session took place in-person and proceeded as follows. Participants first provided informed consent and were briefed about the setup. Then they were assigned to one of the two groups to decide interface order. Participants in first group used provenance-graph for the first alert and table interface for the second alert. The second group does the vice-versa, using table interface for the first alert and provenance-graph interface for the second. Participants spent 37 minutes on average on the first alert, and 40 minutes on the second. There were 27 and 31 ground-truth artifacts in alert #1 and alert #2 respectively. For each alert, the participant used the assigned interface to investigate the related events, and answered the corresponding set of nine assessment questions. We logged actions and submission timestamps throughout. The session ended after both alerts were completed.

## 2.5 Order and Learning Effects

A within-subjects design needs fewer participants for the same number of observations, since each participant contributes data

under both interfaces. This efficiency comes with two threats to internal validity that a between-subjects design would not have. We discuss these issues briefly along with the steps taken to mitigate them.

The first is *carry-over or order effect* [10]. Because every participant experiences both interfaces, whichever one they use second may result in better or worse performance. This could simply happen due to practice or fatigue, independent of the interface itself. We mitigate this with counterbalancing where participants are split into two groups that experience the two interfaces in opposite order. The two groups are Group 1 (provenance-graph  $\rightarrow$  table) and Group 2 (table  $\rightarrow$  provenance-graph). This design averages out any advantage from seeing an interface first or second, isolating the effect of the interface itself.

The second is *practice or learning effect* [10]. If a participant investigated the same alert twice, prior knowledge from the first pass could inflate performance on the second, regardless of which interface was used. We mitigate this by assigning a different alert to each interface condition, so no participant investigates the same attack scenario twice. This means each participant’s performance in the two interfaces of provenance-graph and table come from different alerts. These two alerts are chosen such that they are comparable in difficulty and still involve different set of system entities in the events. We do this to make sure that any learning from the first alert don’t carry forward to the next one.

## 2.6 ACE Platform

We built ACE (Analyst-Centered Evaluation) to run the study, rather than adapting an existing Security Information and Event Management (SIEM) or log-analysis tool. This was done primarily because existing tools are built around a specific query language or workflow. Hence, participants with prior experience in that tool or query syntax may have an advantage independent of whether they were using a provenance graph or table interface. We needed both of the interfaces to expose exactly the same underlying provenance data.

ACE presents each alert through the two interfaces over the same provenance data. The *table* interface lists the events in a flat row-column structure with each row expandable to show full event detail. The *provenance graph* interface renders the same events as a tree, with connected nodes revealed one hop at a time in the left-to-right direction. Similar to expandable rows in table interface, the provenance graph interface also features a sidebar which shows node detail on selection. Both interfaces support the same filtering options, so participants could narrow down events by free-text search, time range, and operation type regardless of which interface they were using.

## 3 Evaluation

We test our hypotheses H1 and H2 against the seven metrics defined in Section 2.3: three quality metrics of precision, recall, and F1 under H1, and four efficiency metrics of question coverage, artifact coverage, time per question, and time per artifact under H2. We refer to these two groups as the RQ1 family and the RQ2 family. Table 3 summarizes the test statistic, effect size, and significance for all seven metrics.

**Table 2: Per-participant question and artifact coverage, precision, and recall by interface, with difference (Diff) calculated for each metric as  $metric_{prov.graph} - metric_{table}$ . For the column ‘Group’, G-T means participant used provenance graph interface for first alert and table for second alert. T-G means the vice-versa. Other metrics are excluded due to space constraints.**

| Participant | Group | Question Coverage |       |      | Artifact Coverage |       |        | Precision |       |        | Recall |       |        |
|-------------|-------|-------------------|-------|------|-------------------|-------|--------|-----------|-------|--------|--------|-------|--------|
|             |       | Graph             | Table | Diff | Graph             | Table | Diff   | Graph     | Table | Diff   | Graph  | Table | Diff   |
| A           | G-T   | 5                 | 4     | 1    | 0.444             | 0.065 | 0.380  | 0.800     | 0.500 | 0.300  | 0.462  | 0.250 | 0.212  |
| B           | T-G   | 6                 | 4     | 2    | 0.226             | 0.074 | 0.152  | 0.583     | 0.667 | -0.083 | 0.250  | 0.143 | 0.107  |
| C           | T-G   | 8                 | 5     | 3    | 0.871             | 0.556 | 0.315  | 0.964     | 1.000 | -0.036 | 0.871  | 0.577 | 0.294  |
| D           | G-T   | 7                 | 6     | 1    | 0.296             | 0.032 | 0.264  | 0.800     | 0.333 | 0.467  | 0.296  | 0.100 | 0.196  |
| E           | T-G   | 9                 | 4     | 5    | 0.323             | 0.296 | 0.026  | 0.625     | 0.800 | -0.175 | 0.323  | 0.571 | -0.249 |
| F           | T-G   | 8                 | 6     | 2    | 0.226             | 0.333 | -0.108 | 0.304     | 0.562 | -0.258 | 0.241  | 0.346 | -0.105 |
| G           | T-G   | 8                 | 7     | 1    | 0.323             | 0.074 | 0.249  | 0.909     | 0.250 | 0.659  | 0.323  | 0.074 | 0.249  |
| H           | T-G   | 5                 | 2     | 3    | 0.194             | 0.037 | 0.157  | 0.667     | 0.083 | 0.583  | 0.667  | 0.091 | 0.576  |
| I           | G-T   | 4                 | 2     | 2    | 0.111             | 0.065 | 0.047  | 0.500     | 0.667 | -0.167 | 0.214  | 0.400 | -0.186 |
| J           | T-G   | 9                 | 3     | 6    | 0.226             | 0.259 | -0.033 | 0.636     | 1.000 | -0.364 | 0.226  | 0.538 | -0.313 |
| K           | T-G   | 9                 | 3     | 6    | 0.548             | 0.148 | 0.400  | 1.000     | 1.000 | 0.000  | 0.548  | 0.308 | 0.241  |
| L           | G-T   | 7                 | 6     | 1    | 0.074             | 0.129 | -0.055 | 0.154     | 0.333 | -0.179 | 0.083  | 0.400 | -0.317 |
| M           | G-T   | 2                 | 4     | -2   | 0.074             | 0.032 | 0.042  | 1.000     | 0.250 | 0.750  | 0.182  | 0.125 | 0.057  |
| N           | G-T   | 9                 | 9     | 0    | 0.556             | 0.226 | 0.330  | 0.682     | 0.500 | 0.182  | 0.556  | 0.226 | 0.330  |
| O           | T-G   | 8                 | 7     | 1    | 0.710             | 0.111 | 0.599  | 0.957     | 0.300 | 0.657  | 0.733  | 0.115 | 0.618  |
| P           | T-G   | 4                 | 1     | 3    | 0.226             | 0.037 | 0.189  | 0.700     | 1.000 | -0.300 | 0.875  | 1.000 | -0.125 |

**Table 3: Wilcoxon signed-rank test on the differences for all seven RQ1/RQ2 metrics. Holm-Bonferroni correction is applied separately within each RQ family ( $k = 3$  for RQ1,  $k = 4$  for RQ2). Effect size  $r = |z|/\sqrt{N}$ . Rank and adjusted  $\alpha$  are ordered by the exact  $p$ -value.**

| RQ  | Measure           | $W$  | $p$ (exact)   | $p$ (approx) | Effect size $r$ | Rank | Adj. $\alpha$ | Sig.       |
|-----|-------------------|------|---------------|--------------|-----------------|------|---------------|------------|
| RQ1 | Recall            | 45.5 | 0.2446        | 0.2446       | 0.291           | 1/3  | 0.0167        | No         |
| RQ1 | Precision         | 40.5 | 0.2680        | 0.2680       | 0.277           | 2/3  | 0.025         | No         |
| RQ1 | F1                | 46.0 | 0.2744        | 0.2553       | 0.284           | 3/3  | 0.050         | No         |
| RQ2 | Question coverage | 7.5  | <b>0.0027</b> | 0.0027       | 0.751           | 1/4  | 0.0125        | <b>Yes</b> |
| RQ2 | Artifact coverage | 13.0 | <b>0.0027</b> | 0.0045       | 0.711           | 2/4  | 0.0167        | <b>Yes</b> |
| RQ2 | Time per artifact | 17.0 | <b>0.0063</b> | 0.0084       | 0.659           | 3/4  | 0.025         | <b>Yes</b> |
| RQ2 | Time per question | 26.0 | <b>0.0290</b> | 0.0299       | 0.543           | 4/4  | 0.050         | <b>Yes</b> |

### 3.1 Analysis plan

The within-subjects design gives one paired observation per participant for each metric: a value for provenance graph interface and a value for table interface. For example, in table 2, participant B completed 6 of 9 questions using the provenance graph interface and 4 of 9 questions using the table interface, contributing the pair (6, 4) for metric of question coverage. For each metric, we compute the paired difference for every participant e.g. the pair (6, 4) gives us the difference of 2 i.e.  $(ques.coverage_{prov.graph} - ques.coverage_{table})$ . These differences did not meet the assumptions of normality and showed skewed distributions, so we use the Wilcoxon signed-rank test [27, 35], a non-parametric test for paired samples, as our primary test for every metric under H1 and H2.

Running seven tests across H1 and H2 raises the risk of the family-wise error rate i.e. a false positive occurring by chance. We

control for this with Holm-Bonferroni correction [1, 13] to the significance threshold  $\alpha = 0.05$ , applied separately within each family: the three RQ1 metrics under H1, and the four RQ2 metrics under H2. Order effects are assessed separately in Section 3.4, as they are a property of the study design rather than of any individual metric. We report results for each metric below and evaluate whether performance differs significantly between the provenance-graph and table interfaces.

### 3.2 Quality in Attack Artifact Discovery (RQ1)

Precision, recall, and F1-score are computed over the questions a participant attempted, classifying each reported artifact as TP (valid attack artifact), FP (artifact not part of the attack), or FN (missed attack artifact). Together they capture the quality of what

participants found, independent of how much ground they covered in investigation.

**Precision** was higher for participants using the provenance graph (median 0.691, IQR [0.614, 0.921]) than the table (median 0.531, IQR [0.325, 0.850]). A paired Wilcoxon signed-rank test shows the difference is not significant ( $W = 40.5$ ,  $N = 16$ ,  $p = 0.268$ ,  $r = 0.277$ ), tested against a Holm-Bonferroni adjusted alpha of 0.025.

**Recall** showed the same pattern. Participants using the provenance graph reported higher recall (median 0.323, IQR [0.237, 0.584]) than using the table (median 0.279, IQR [0.122, 0.434]), but the difference is not significant ( $W = 45.5$ ,  $N = 16$ ,  $p = 0.245$ ,  $r = 0.291$ ) against an adjusted alpha of 0.017.

**F1-score** combines both. Participants using the provenance graph reported higher F1-score (median 0.454, IQR [0.327, 0.677]) than table participants (median 0.348, IQR [0.167, 0.542]), again without reaching significance ( $W = 46.0$ ,  $N = 16$ ,  $p = 0.255$ ,  $r = 0.284$ ) against an adjusted alpha of 0.050.

Hence *quality* was descriptively higher for participants using the provenance graph with small-to-moderate ( $r = 0.28$ – $0.29$ ) effect sizes across all three metrics. However, none of the differences reached significance.

### 3.3 Efficiency in Attack Artifact Discovery (RQ2)

The *quality* metrics showed a consistent but non-significant trend. In contrast, the results for *efficiency* are more decisive.

**Question coverage** was higher for participants using the provenance graph (median 7.5 of 9, IQR [5.0, 8.25]) than the table (median 4.0 of 9, IQR [3.0, 6.0]). The difference is significant ( $W = 7.5$ ,  $N = 16$ ,  $p = 0.0027$ ,  $r = 0.751$ ) against the adjusted alpha of 0.0125.

**Artifact coverage** follows the same pattern. Participants using the provenance graph reported a larger share of the total ground-truth artifacts (median 0.261, IQR [0.218, 0.470]) than table participants (median 0.092, IQR [0.058, 0.234]), also significant ( $W = 13.0$ ,  $N = 16$ ,  $p = 0.0045$ ,  $r = 0.711$ ) against an adjusted alpha of 0.017. The upper quartile in the table condition (0.234) falls below the median in the provenance-graph condition (0.261). This means that a participant at the 75th percentile with the table interface still typically found less evidence than a median performer with the provenance graph interface.

**Time per artifact** was lower for the graph (median 247.8s, IQR [148.9, 333.2]) than the table (median 557.6s, IQR [293.2, 1192.9]), a significant difference ( $W = 17.0$ ,  $N = 16$ ,  $p = 0.0084$ ,  $r = 0.659$ ) against an adjusted alpha of 0.025.

**Time per question** was also lower for the graph (median 295.8s, IQR [253.6, 428.8]) than the table (median 461.2s, IQR [310.4, 560.7]). This is the weakest of the four RQ2 results but still clears its adjusted alpha of 0.050 ( $W = 26.0$ ,  $N = 16$ ,  $p = 0.0299$ ,  $r = 0.543$ ).

All four RQ2 metrics point in the same direction. Participants using the provenance graph covered more of the alert, found more of the ground-truth artifacts, and did so faster per question and per correctly identified artifact. Given how strong and consistent this pattern is, we now investigate if it is genuinely about the interface, or whether it is at least partly explained by which interface a participant happened to use first.

### 3.4 Order Effects

Counterbalancing splits participants into two groups that use the interfaces in opposite order, which should cancel out any advantage from going first or second when averaged across the participants. But counterbalancing only guarantees this on average and it does not tell us whether the size of the provenance graph's advantage was the same for participants who saw it first as for those who saw it second. We check this for each of the seven metrics. We compare the difference i.e.  $metric_{prov.graph} - metric_{table}$  between Group A (provenance-graph  $\rightarrow$  table,  $n = 6$ ) and Group B (table  $\rightarrow$  provenance-graph,  $n = 10$ ) with a Mann-Whitney  $U$  test [24, 27].

Table 4 summarizes the results. None of the three RQ1 metrics showed an order effect: precision ( $U = 38.0$ ,  $p = 0.4278$ ), recall ( $U = 24.0$ ,  $p = 0.5622$ ), or F1 ( $U = 30.0$ ,  $p = 1.0000$ ). Two of the four RQ2 measures did.

**Question coverage** showed a significant order effect ( $U = 6.0$ ,  $p = 0.0092$ ): Group A completed 0.5 more questions on average (median 1.0, SD 1.38) using the provenance graph, while Group B completed 3.2 more (median 3.0, SD 1.87). Every participant in Group B attempted at least one more question while using the provenance graph (min 1, max 6). **Time per question** showed the same pattern ( $U = 57.0$ ,  $p = 0.0017$ ): Group A was on average 106.5s slower per question with the provenance graph (median  $-7.3$ s), while Group B was 352.7s faster (median  $-232.2$ s). Participants who used the provenance graph second, after already having used the table, attempted more questions and moved faster per question than those who saw it first. **Artifact coverage** ( $U = 30.0$ ,  $p = 1.0000$ ) and **time per artifact** ( $U = 34.0$ ,  $p = 0.7128$ ) showed no such effect.

To summarize, two of four RQ2 metrics (Question coverage and Time per question) showed evidence of order effects while the other two (Artifact coverage and Time per artifact) did not.

### 3.5 Weighing the Evidence

Not all seven results carry the same weight, so we rank them by three criteria together 1) whether the difference is significant after applying correction 2) how large the effect size is, and 3) whether Section 3.4 found an order effect that complicates attributing the difference to the interface alone.

**Artifact coverage and time per artifact** meet all three criteria: both are significant after correction, both carry large effect sizes ( $r = 0.711$  and  $r = 0.659$ ), and neither shows an order effect. These two metrics have the cleanest evidence in the study. Using the provenance graph increased the share of ground-truth artifacts participants found, and reduced the time needed per correctly identified artifact.

**Question coverage and time per question** are also significant, with large-to-moderate effect sizes ( $r = 0.751$  and  $r = 0.543$ ), but both show a significant order effect. The advantage was larger for participants who used the provenance graph second rather than first. We still state that the provenance graph increased question coverage and reduced time per question, but the size of this effect is entangled with practice on the assessment itself, not the interface alone. This means participants likely benefited from becoming more familiar with the task itself, and this benefit was compounded when the provenance graph interface came second in Group B.

**Table 4: Order effect check across all RQ1 and RQ2 metrics (Mann-Whitney  $U$  on their differences between the two independent groups, Group A = provenance-graph-then-table,  $n = 6$ ; Group B = table-then-provenance-graph,  $n = 10$ ).**

| RQ  | Measure               | $U$  | $p$           | Order effect | Group A mean $\pm$ SD | Group B mean $\pm$ SD |
|-----|-----------------------|------|---------------|--------------|-----------------------|-----------------------|
| RQ1 | Precision             | 38.0 | 0.4278        | No           | 0.226 $\pm$ 0.363     | 0.068 $\pm$ 0.406     |
| RQ1 | Recall                | 24.0 | 0.5622        | No           | 0.049 $\pm$ 0.252     | 0.129 $\pm$ 0.325     |
| RQ1 | F1                    | 30.0 | 1.0000        | No           | 0.087 $\pm$ 0.250     | 0.115 $\pm$ 0.357     |
| RQ2 | Question coverage     | 6.0  | <b>0.0092</b> | <b>Yes</b>   | 0.500 $\pm$ 1.378     | 3.200 $\pm$ 1.874     |
| RQ2 | Artifact coverage     | 30.0 | 1.0000        | No           | 0.168 $\pm$ 0.179     | 0.195 $\pm$ 0.210     |
| RQ2 | Time per question (s) | 57.0 | <b>0.0017</b> | <b>Yes</b>   | 106.49 $\pm$ 238.88   | -352.70 $\pm$ 421.43  |
| RQ2 | Time per artifact (s) | 34.0 | 0.7128        | No           | -393.40 $\pm$ 761.01  | -556.51 $\pm$ 571.06  |

**Precision, recall, and F1** favor the provenance graph in direction but not significance, with small-to-moderate effect sizes ( $r = 0.28$ – $0.29$ ). At  $N = 16$ , the study likely lacked the statistical power to detect a difference of this size, even if one exists.

## 4 Discussion

Table 5 summarizes our findings. Provenance graphs helped participants cover more attack surface and collect artifacts faster than tables. The quality of what participants collected, however, did not change significantly between these interfaces.

### 4.1 Implications

We set out with two hypotheses about provenance graphs for attack artifact discovery. We discuss our findings against each of these hypothesis in turn.

**H1.** Provenance graphs will increase quality in artifact discovery relative to tables, measured as precision, recall, and F1.

*Comparable quality.* Precision did not differ significantly between interfaces ( $p = 0.268$ ), nor did recall ( $p = 0.245$ ) or F1-score ( $p = 0.274$ ). It was our intuition that provenance graphs would help participants visualize attack steps more clearly than a table, which would result in better quality of collected attack artifacts. However, that was not the case and participants using table interface fared comparably. Figure 3 plots these metrics per participant and the spread of data shows participants performed overall better with the provenance graph interface in comparison to using table, though not significantly.

**Takeway:** Our study lacked the statistical power required to detect fine-grained performance difference with respect to the quality of collected attack artifacts. The bootstrap 95% confidence intervals show a wide range in median differences of precision ( $-0.018$ , CI:  $[-0.175, 0.467]$ ), recall ( $0.152$ , CI:  $[-0.125, 0.253]$ ) and F1 ( $0.162$ , CI:  $[-0.200, 0.278]$ ). While not conclusive, these differences still favor provenance graph and we see in figure 3 that 25th percentile of participants using provenance graph interface performed comparably to the median of those using table interface. We suggest that future works perform similar studies with larger cohorts to detect these differences.

**H2.** Provenance graphs will increase efficiency relative to tables, measured as question coverage, artifact coverage, time per question, and time per artifact.

*Same time, more output.* Total session time did not differ significantly between interfaces (median 2087s graph vs. 1973s table,  $p = 0.11$ ). Within that time, participants using the provenance graph completed 83% of assessment questions (7.5/9) against 44% (4/9) with the table, found 2.8 $\times$  more ground-truth artifacts, and spent 55% less time doing so.

**Takeway:** Provenance graphs helped participants cover more attack surface and find more artifacts without taking more time to do it. For a security team, this reads as evidence that provenance graphs let an operator collect more evidence in the same investigation window.

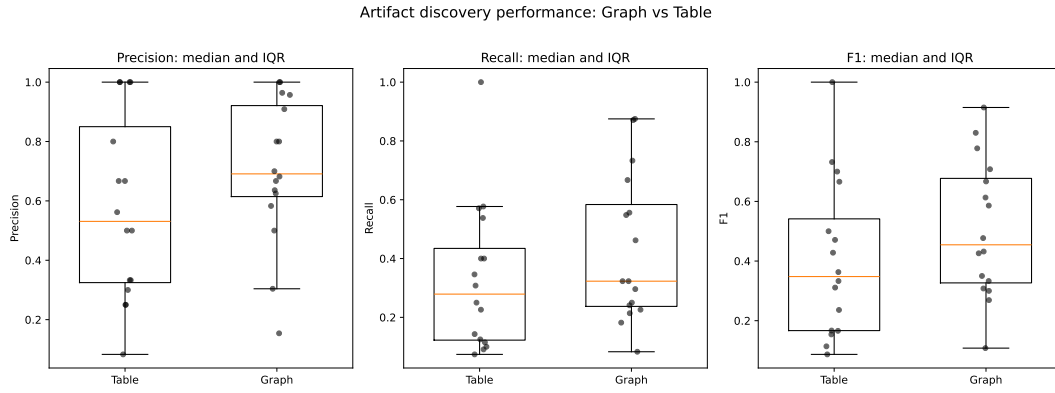
### 4.2 Provenance graphs for alert investigation

**Visualizing provenance graphs is hard.** We borrow this phrase from authors in their work [9] who worked closely with security practitioners to validate the commercial potential of their provenance-based startup. Similar to our work, they prepared an interactive demo and identified *node placement* as one of the challenges in visualizing provenance graphs for alerts and the constituent events. For our platform ACE, our first implementation used a force-directed layout, where nodes repel each other and edges pull connected nodes together. Security operators expect a left-to-right sense of event progression [9], which a force-directed layout does not provide because it has no notion of temporal or causal direction. Eventually, we replaced it with a layout that fixes the starting alert node in place and positions connected nodes to its right, revealing each hop of the neighborhood on demand rather than all at once. Even with this change, some participants found the resulting graphs hard to read. For instance in our first alert scenario, a `/usr/bin/firefox` process read multiple `/proc/*/stat` files 310 times, including `/etc/passwd` (322 reads). This is genuinely useful evidence, but rendering each read operation with `/proc/*/stat` as its own node produced vertical-tall, repetitive visualizations. As a result, a few participants missed that the same `firefox` process also read `/proc/*/cmdline` and `/proc/*/status`, which were buried under volume and not absent from the provenance graph.

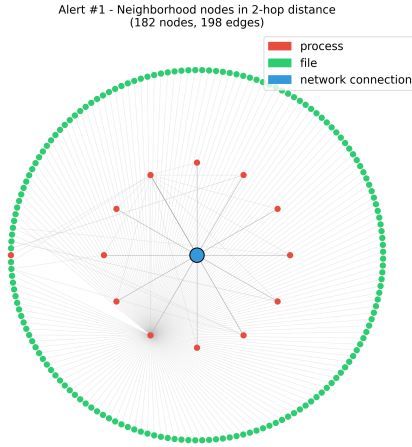
**Provenance data is detailed and noisy.** Analysts do not work with the level of granularity offered by provenance data [9]. During

**Table 5: Provenance graph vs. table performance across all seven RQ1/RQ2 metrics. Ratios are provenance-graph÷table throughout. For the two time-based metrics, a ratio below 1 indicates the provenance graph interface was faster. The last column says whether the difference between interfaces was significant or not for that metric. An asterisk (\*) means an order effect was likely present.**

| Metric                 | RQ  | Mean G / T    | Ratio (Mean) | Median G / T  | Ratio (Median) | What it means                     | Sig. |
|------------------------|-----|---------------|--------------|---------------|----------------|-----------------------------------|------|
| Precision              | RQ1 | 0.705 / 0.578 | 1.22×        | 0.691 / 0.531 | 1.30×          | Precision was 1.30× table's       | No   |
| Recall                 | RQ1 | 0.428 / 0.329 | 1.30×        | 0.323 / 0.279 | 1.16×          | Recall was 1.16× table's          | No   |
| F1                     | RQ1 | 0.506 / 0.402 | 1.26×        | 0.454 / 0.348 | 1.31×          | F1 was 1.31× table's              | No   |
| Question coverage (/9) | RQ2 | 6.75 / 4.56   | 1.48×        | 7.5 / 4.0     | 1.88×          | Attempted 1.88× as many questions | Yes* |
| Artifact coverage      | RQ2 | 0.339 / 0.155 | 2.19×        | 0.261 / 0.092 | 2.82×          | Recovered 2.82× as many artifacts | Yes  |
| Time per question (s)  | RQ2 | 359.4 / 539.9 | 0.66×        | 295.8 / 461.2 | 0.64×          | Took 36% less time per question   | Yes* |
| Time per artifact (s)  | RQ2 | 341.3 / 836.7 | 0.41×        | 247.8 / 557.6 | 0.44×          | 55% less time per artifact        | Yes  |



**Figure 3: Comparing spread of Precision, Recall and F1 metrics for provenance graph and table**  
The horizontal line marks the median. Jittered points show individual participant values.



**Figure 4: The center node is the starting point in alert investigation and represents a network connection to the attacker's IP address. The surrounding nodes represents the /usr/bin/firefox processes including forks/clones that reads multiple files depicted as green nodes. Only one of these files is a malicious payload.**

the experiment session, we also expected participants to ask us questions about the event types since they did not receive any special training or preparation lessons. To our surprise, the provenance graphs were easy to interpret for them. This resonates with the findings in [7] that provenance graphs consisting of low-level system events are interpretable, as long as they are concise. In our case, we did not filter out any events, however we merged events between the same entities to show as a single edge in the provenance graphs. This improved readability but the resulting provenance graph still contained hundreds of nodes i.e. system entities in the events. For example, figure 4 shows the nodes in only the 2-hop neighborhood in first alert scenario. The blue-colored center node represents a network connection to 146.153.68.151:80 and is the starting point in alert investigation. The process /usr/bin/firefox including its child processes (denoted with red nodes) communicates with this IP address and reads multiple files. These files are visualized in green in the figure 4. Out of total 168 nodes representing the files, there is only one /home/admin/clean file which is the malicious payload written by usr/bin/firefox. Investigating this payload file further allows us to see the complete attack chain. Figure 5 shows this attack chain in our ACE tool. We note that this provenance graph is concise as it only contains attack-relevant events. Although a security operator needs to explore up to 6-hops, the provenance

graph made from raw audit logs still contains a significantly lower amount of nodes/edges that captures the complete attack.

Based on these observations and feedback from participants, we suggest two concrete directions. First, aggressive pruning techniques will help reduce the information security operators will see. For example, log-reduction techniques [15] can collapse repeated, structurally similar events (such as the 310 /proc/\*/stat reads) into a single summarized node. For example, in figure 6 repeated activity with similar files can be summarized with regular expression as proposed in the work LogApprox [25]. The security operators should be able to expand such summarized nodes on request. This way, an investigation will preserve the underlying evidence without forcing the user to scroll past it node by node. Second, the security operator should be able to opt for readability on demand. This means the number of nodes can be capped in every hop in the provenance graph, with the limit adjustable by the security operator.

**Lack of human studies.** We echo the authors' call for more controlled experiments [15] that involves practitioners and stakeholders interested in provenance-based systems. To the best of our knowledge, our study is the first to evaluate the effects of provenance graph in alert investigation. Just like other studies, our results also come from one single session. Security operators use mental models that are learned from prior experience and developed over time. Adopting provenance graphs successfully means security operators include them in their workflow. Hence future work can investigate and identify the exact stages/parts of the workflow that stands to benefit the most from the causal structure of provenance graphs. We anticipate that conducting long-horizon studies could be a way forward. Understanding the long-term performance and behavior changes caused by these provenance-based solutions can provide fine-grained insights into the needs of security operators.

**Industry-perspective on provenance graphs.** Industry agrees that provenance-based EDR solutions offer better interpretability [7] and hence, are better than traditional EDRs. These solutions use provenance graph to link related alerts and facilitates root cause analysis while traditional EDR alerts are isolated in nature. However, using provenance graph asks operators to shift away from the tabular, query-driven workflows they already know. Better interpretability do not automatically translate into better response outcomes. In their evaluation of PIDS in industrial scenarios [36], authors reported high investigation overhead and shared that *the fine-grained detection results in provenance graphs do not align fully with the requirements of attack investigation*. We observe the same in another work SYSARMOR [20] where they integrated a PIDS in their organization EDR system. Then, over the period of one week, they conducted an APT-style attack simulation in their environment and the generated alerts were visualized in concise provenance graphs with Tactics, Techniques, and Procedures (TTPs) techniques. Although the proposed system successfully detected the attack stages, the authors pointed out the gap between detection and response— security operators need more actionable insights to respond to the detected threats in real-world scenario.

The authors in the survey [7] of the gap between academic research and commercial deployment of EDR systems identified the *cost of interpretation* as one of the major bottlenecks in adopting provenance-based systems. However, they agreed to use the number

of nodes in the provenance graph as the metric for this cost. We believe that it is more nuanced because a provenance graph might be smaller in size, but it can still not be actionable. On the other hand, a larger provenance graph with actionable next steps could be easier to interpret and effective for the security operator. This suggests that provenance graphs might offer richer causal representation but exploiting/leveraging it requires changes on two fronts: 1) Security operators shift/change their existing mental models of conducting investigations. 2) Provenance graphs translate the PIDS alerts into actionable, remediation steps.

### 4.3 Limitations

We list four aspects of our design that limit how far these results generalize.

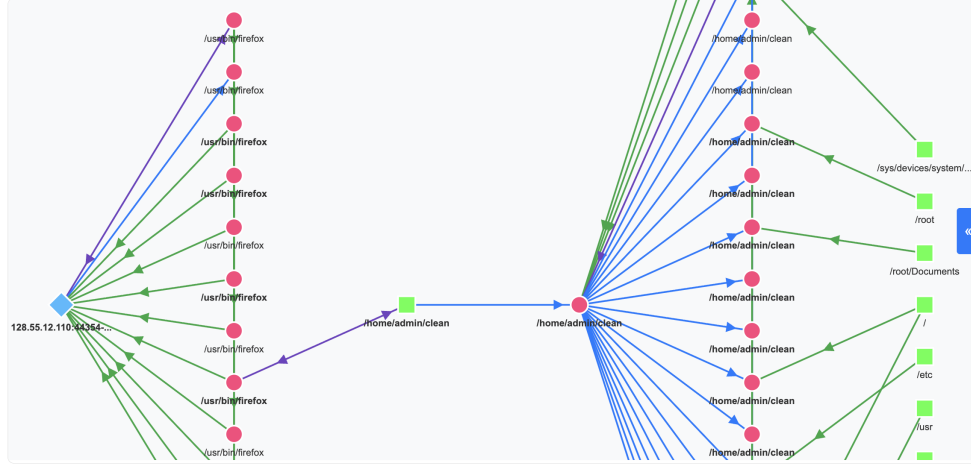
**Novice participants.** The participants in our study were graduate students with cybersecurity coursework, not employed tier-1 security operators working live alerts under real time pressure. Their triage strategies, pace, and tolerance for ambiguity likely differ from an operational SOC analyst's. Recruiting practicing analysts for a multi-hour lab study carries a cost few SOC's can absorb, which is why we ran this study with students as a course assignment. Validating these findings with practitioners remains necessary before treating them as SOC-ready guidance.

**Single attack scenario.** Each participant investigated each alert under only one interface. And both of these alerts come from the DARPA THEIA Engagement-3 dataset which is generated from one simulated APT campaign. We selected the two alerts to be comparable in difficulty while still differing in the system entities involved. Even so, both come from one attack campaign. Provenance graphs and event structures differ across attack types, and this can change how difficult an attack is to interpret. Therefore, a broader claim about provenance graphs across attack types and datasets needs more scenarios than the two that were included in the study.

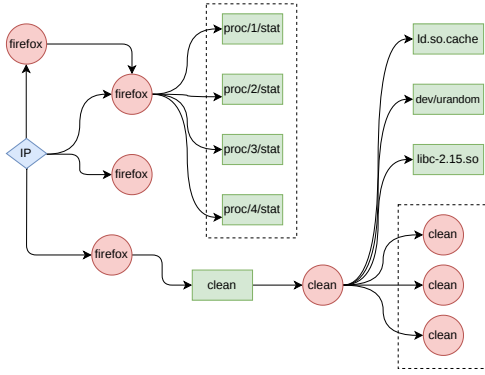
**ACE platform.** Every result in this paper was collected on a platform we built ourselves. Our conclusions describe the specific hop-anchored, incrementally-revealed layout implemented in ACE, not provenance-graph visualization as an abstract category. A different layout, a different set of filters, or a different table design could plausibly narrow or widen the gap we observed. However, to the best of our knowledge, our study is the first that uses the left-to-right layout of provenance graph which aligns closely with the expectations of security practitioners. We also plan to implement pruning techniques such as LogApprox [25] in ACE to make the provenance graphs more practical and compact in alert investigations.

**Low sample size.** At  $N = 16$ , the study has limited statistical power, which is the most direct explanation for why RQ1's quality metrics showed moderate effect sizes ( $r = 0.28-0.29$ ) without reaching significance. Nevertheless, the quality did not degrade under the larger workload, because participants using the provenance graph interface attempted nearly twice as many questions and recovered nearly three times as many artifacts as compared to using table interface.

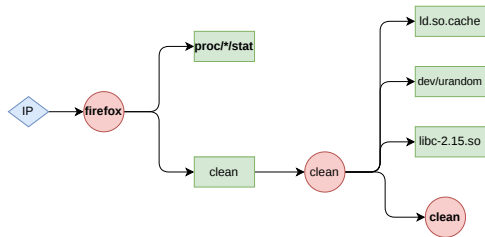
**Cost of switching.** Security operators are accustomed to tabular, query-based investigation and including provenance graph asks them to change this habit. Our work indicates evidence that this



**Figure 5: Provenance graph of attack scenario in alert #1 in our ACE platform. After communicating with the attacker’s IP address, one of the `/usr/bin/firefox` processes writes a malicious payload to `/home/admin/clean` which later executes with root privileges.**



**(a) Lossless Provenance Graph**



**(b) LogApprox’s Attack-Preserving Provenance Graph**

**Figure 6: Simplified provenance graphs of the initial attack steps in alert #1 (a) depicts the raw audit logs while (b) shows the provenance graph produced after technique like LogApprox [25] is applied.**

switch could result in better artifact coverage and speed. However, we don’t measure the cost of switching or the training time.

#### 4.4 Ethical Considerations

The study was approved by the ethics review board in our institution. We strictly follow the research data management policy for data storage, sharing and processing. We acquired explicit consent from the participants.

#### 5 Related Work

Provenance-based systems produce an alert as a subgraph of the entities involved and the events surrounding them. This subgraph gives the analyst causal context, however, the resulting alert graphs can also become hard to inspect during an investigation. For e.g. these provenance graphs will contain benign events that are not part of the attack necessarily. Moreover, the provenance graph also grows quickly with the amount of activity it captures. Hence, a primary focus of provenance research for alert investigation is tackling the twin challenges of dependency explosion [19] and information overload [17, 33].

**Provenance-based Investigation Systems.** Systems such as NoDoze [33] and WATSON [38] focus on automated alert triage and behavior abstraction to reduce workload of the security operator. NoDoze ranks alerts and reduces false positives while Watson groups low-level audit events into a smaller set of high-level behaviors. To aid semantic interpretation, OMEGALOG [12] bridges system audit logs with application-layer events, while RAP SHEET [11] and TREC [23] map raw causal chains directly to high-level MITRE ATT&CK tactics and techniques. Other techniques leverage structural and causal analysis to extract critical subgraphs (DEPIIMPACT [8]), partition graphs into community summaries (DEPCOMM [37]), or recover attack story from system audit logs (ATLAS [2]). RAPID [22] provides a real-time system for alert prioritization and investigation. The implicit assumption here is, that concise or semantically

**Table 6: Attack-investigation systems and their evaluation of claimed analyst benefit.** Papers in **bold** included end users in their evaluation; for RAPID and ProvTalk this was limited to opinion or perception feedback, without task-based performance measurement against ground truth.

| Paper                                                  | Venue     | Claimed benefit to analyst               | Evaluation criteria                                                                                     |
|--------------------------------------------------------|-----------|------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <i>Provenance-based investigation systems</i>          |           |                                          |                                                                                                         |
| NoDoze [33]                                            | NDSS'19   | Reduce alert fatigue; concise graphs     | Alert ranking (ROC/AUC); graph reduction; estimated hours saved                                         |
| OmegaLog [12]                                          | NDSS'20   | Semantically rich investigation          | Graph conciseness; applicability; overhead                                                              |
| RapSheet [11]                                          | S&P'20    | Compact visuals accelerate investigation | Tactical Provenance Graphs ranking (ROC); response time; log reduction                                  |
| Watson [38]                                            | NDSS'21   | Reduce analysis workload                 | Abstraction accuracy (F1); events-to-inspect count                                                      |
| ATLAS [2]                                              | USENIX'21 | Recover attack stories                   | Attack-entity precision/recall                                                                          |
| DepImpact [8]                                          | USENIX'22 | Focus analysts on critical subgraph      | Graph reduction; entry-node identification                                                              |
| DepComm [37]                                           | S&P'22    | Summaries reduce inspection effort       | Community detection (F1); compression rate                                                              |
| <b>RAPID</b> [22]                                      | ACSAC'22  | Reduce investigation workload            | Space/time efficiency; analyst-feedback survey                                                          |
| <b>ProvTalk</b> [31]                                   | NDSS'22   | Interpretable root-cause analysis        | Graph reduction; overhead; perceived interpretability (Likert)                                          |
| ProvG-Searcher [3]                                     | CCS'23    | Efficient threat-behavior search         | Matching accuracy/F1; query latency                                                                     |
| TREC [23]                                              | CCS'24    | ATT&CK semantics for analysts            | Tactic/technique recognition accuracy                                                                   |
| Orthrus [17]                                           | USENIX'25 | Minimize analyst effort                  | Node-level precision/MCC vs. ground truth                                                               |
| <i>Human-subject studies on adjacent analyst tasks</i> |           |                                          |                                                                                                         |
| <b>Kersten et al.</b> [18]                             | SOUPS'23  | —                                        | Tier-1 alert-assessment correctness (process intervention)                                              |
| <b>Jansen et al.</b> [16]                              | WOSOC'24  | —                                        | Rule-creation difficulty (log vs. graph)                                                                |
| <b>Provexa</b> [32]                                    | VLDL'25   | —                                        | Query-language usability (ProvQL vs. SQL/Cypher)                                                        |
| <b>Moosmann et al.</b> [26]                            | DMVA'26   | —                                        | Triage decision explanations                                                                            |
| <b>This work</b>                                       | —         | —                                        | Analyst performance on attack artifact discovery (provenance graph vs. table, within-subjects, $N=16$ ) |

enriched graphs directly translate to superior human analyst performance. However, we rarely see the end-users being involved in the evaluation of such systems. We summarize some of these provenance-based attack investigation systems in table 6. We also include some works that focus on user-studies on tasks similar to attack artifact discovery in this work.

**Evaluation with Human Subjects.** A smaller body of work measures analysts directly, on tasks around the investigation. Kersten et al. [18] show that a structured process improves how correctly tier-1 analysts assess alerts, and Moosmann et al. [26] investigate further to see if such SOC operators can explain their decisions for such assessments. The authors report that creating accurate, evidence-based explanation is still challenging for such security operators. Other work that is more relevant to provenance graphs/analysis is [16]. The authors conduct a study to compare how difficult it is to write graph-based detection rules versus log-based ones. However, their work focuses on effective threat detection and not investigation that follows it. Provexa [32] propose a domain-specific language ProvQL, tailored to system provenance and attack investigation. The authors compare the practicality and usability of their solution to graph-based query language, Cypher and SQL. The goal of their study is to evaluate the effectiveness of their tool for threat investigation and not how the presentation affects the same. Among these works, Kersten et al. [18] and Moosman et al. [26] conducted the user-studies in real-world SOC. While the research community would greatly benefit from similar studies, such experiments are costly with actual SOC operators. In comparison, we conducted our study with Masters students as part of course assignment. Such students represent the junior analysts in SOC and can serve as a reasonable proxy when practitioner access is limited. We recommend this approach to future studies facing similar constraints, provided participants have comparable coursework or background in the relevant domain.

## 6 Conclusion

Provenance graphs are a promising support tool for alert investigation. Compared to tables, they connect causally related events and thereby help the analyst reconstruct and understand the threat activity. This usefulness, however, rests on the claim that better interpretability leads to better investigation outcomes. We test this claim with a user study and examine how provenance graphs affect the *quality* and *efficiency* of security operators in an alert investigation. Our results show that the same underlying data, shown as a provenance graph instead of a table, improved efficiency— participants recovered nearly 3× more attack artifacts and spent 55% less time per artifact. The quality of collected attack artifacts however, showed no significant difference— participants with the provenance graph were able to investigate more of the attack scenario, but the attack artifacts they found were not better than what table users shared. The benefit of provenance graphs is therefore not uniform across the investigation. If the task is gathering evidence, provenance graphs helps in covering more parts of the attack and moving forward in alert investigation. If the task is judging the evidence, provenance graphs did not provide any substantial advantage.

We anticipate that, to truly leverage the advantage provenance graphs provide, we must identify which investigation tasks they are suited for, rather than proposing them for the investigation as a whole. Future studies can extend our task-level comparison to the stages we did not measure, such as the decision to escalate an alert, and attacks beyond the two scenarios studied here. Our platform and study design are available to support such studies.

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## References

- [1] Hervé Abdi. 2010. Holm's sequential Bonferroni procedure. *Encyclopedia of research design* 1, 8 (2010), 1–8.
- [2] Abdulullah Alsaheel, Yuhong Nan, Shiqing Ma, Le Yu, Gregory Walkup, Z. Berkay Celik, Xiangyu Zhang, and Dongyan Xu. 2021. ATLAS: A Sequence-based Learning Approach for Attack Investigation. In *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 3005–3022. <https://www.usenix.org/conference/usenixsecurity21/presentation/alsaheel>
- [3] Enes Altinisik, Fatih Deniz, and Hüsrev Taha Sencar. 2023. Provg-searcher: A graph representation learning approach for efficient provenance graph search. In *Proceedings of the 2023 ACM SIGSAC conference on computer and communications security*. 2247–2261.
- [4] Tristan Bilot, Baoxiang Jiang, Zefeng Li, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, and Thomas Pasquier. 2025. Sometimes Simpler is Better: A Comprehensive Analysis of State-of-the-Art Provenance-Based Intrusion Detection Systems. In *Security Symposium (USENIX Sec'25)*. USENIX.
- [5] DARPA. 2020. Transparent Computing Engagement 3 and 5 Data Release. <https://github.com/darpa-i2o/Transparent-Computing>.
- [6] DARPA Five Directions. 2020. Operationally Transparent Cyber (OpTC) Data Release. <https://github.com/FiveDirections/OpTC-data>.
- [7] Feng Dong, Shaofei Li, Peng Jiang, Ding Li, Haoyu Wang, Liangyi Huang, Xusheng Xiao, Jiedong Chen, Xiapu Luo, Yao Guo, et al. 2023. Are we there yet? an industrial viewpoint on provenance-based endpoint detection and response tools. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*. 2396–2410.
- [8] Pengcheng Fang, Peng Gao, Changlin Liu, Erman Ayday, Kangkook Jee, Ting Wang, Yanfang Ye, Zhuotao Liu, and Xusheng Xiao. 2022. Back-Propagating System Dependency Impact for Attack Investigation. In *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, Boston, MA, 2461–2478. <https://www.usenix.org/conference/usenixsecurity22/presentation/fang>
- [9] Akul Goyal and Adam Bates. 2026. Lessons Learned through Customer Discovery in a Provenance-based Security Start-Up. In *PRISM Workshop - NDSS 2026*. <https://www.ndss-symposium.org/wp-content/uploads/prism2026-11.pdf>
- [10] Anthony G Greenwald. 1976. Within-subjects designs: To use or not to use? *Psychological Bulletin* 83, 2 (1976), 314.
- [11] Wajih Ul Hassan, Adam Bates, and Daniel Marino. 2020. Tactical provenance analysis for endpoint detection and response systems. In *2020 IEEE symposium on security and privacy (SP)*. IEEE, 1172–1189.
- [12] Wajih Ul Hassan, Mohammad Ali Nouredine, Pubali Datta, and Adam Bates. 2020. Omegalog: High-fidelity attack investigation via transparent multi-layer log analysis. In *Network and distributed system security symposium*.
- [13] Sture Holm. 1979. A simple sequentially rejective multiple test procedure. *Scandinavian journal of statistics* (1979), 65–70.
- [14] Md Nahid Hossain, Sanaz Sheikhi, and R Sekar. 2020. Combating dependence explosion in forensic analysis using alternative tag propagation semantics. In *2020 IEEE symposium on security and privacy (SP)*. IEEE, 1139–1155.
- [15] Muhammad Adil Inam, Yinfang Chen, Akul Goyal, Jason Liu, Jaron Mink, Noor Michael, Sneha Gaur, Adam Bates, and Wajih Ul Hassan. 2023. Sok: History is a vast early warning system: Auditing the provenance of system intrusions. In *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2620–2638.
- [16] Matt Jansen, Rakesh Bobba, and Dave Nevin. 2024. A Comparative Analysis of Difficulty Between Log and Graph-Based Detection Rule Creation. In *Workshop on SOC Operations and Construction (WOSOC) 2024*.
- [17] Baoxiang Jiang, Tristan Bilot, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, Shahrear Iqbal, Xueyuan Han, and Thomas Pasquier. 2025. ORTHRUS: Achieving High Quality of Attribution in Provenance-based Intrusion Detection Systems. In *34th USENIX Security Symposium (USENIX Security 25)*. USENIX Association, Seattle, WA, 7173–7192. <https://www.usenix.org/conference/usenixsecurity25/presentation/jiang-baoxiang>
- [18] Leon Kersten, Tom Mulders, Emmanuele Zamboni, Chris Snijders, and Luca Allodi. 2023. 'Give Me Structure': Synthesis and Evaluation of a (Network) Threat Analysis Process Supporting Tier 1 Investigations in a Security Operation Center. In *Nineteenth Symposium on Usable Privacy and Security (SOUPS 2023)*. USENIX Association, Anaheim, CA, 97–111. <https://www.usenix.org/conference/soups2023/presentation/kersten>
- [19] Kyu Hyung Lee, Xiangyu Zhang, and Dongyan Xu. 2013. High Accuracy Attack Provenance via Binary-based Execution Partition. In *NDSS*, Vol. 16.
- [20] Shaofei Li, Jiandong Jin, Hanlin Jiang, Yi Huang, Yifei Bao, Yuhang Meng, Feng Hong, Zheng Huang, Peng Jiang, and Ding Li. 2026. SYSARMOR: The Practice of Integrating Provenance Analysis into Endpoint Detection and Response Systems. In *PRISM Workshop - NDSS 2026*. <https://www.ndss-symposium.org/wp-content/uploads/prism2026-2.pdf>
- [21] Xiaojing Liao, Kan Yuan, XiaoFeng Wang, Zhou Li, Luyi Xing, and Raheem Beyah. 2016. Acing the ioc game: Toward automatic discovery and analysis of open-source cyber threat intelligence. In *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*. 755–766.
- [22] Yushan Liu, Xiaokui Shu, Yixin Sun, Jiyong Jang, and Prateek Mittal. 2022. Rapid: real-time alert investigation with context-aware prioritization for efficient threat discovery. In *Proceedings of the 38th Annual Computer Security Applications Conference*. 827–840.
- [23] Mingqi Lv, HongZhe Gao, Xuebo Qiu, Tieming Chen, Tiantian Zhu, Jinyin Chen, and Shouling Ji. 2024. TREC: APT tactic/technique recognition via few-shot provenance subgraph learning. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*. 139–152.
- [24] Henry B Mann and Donald R Whitney. 1947. On a test of whether one of two random variables is stochastically larger than the other. *The annals of mathematical statistics* (1947), 50–60.
- [25] Noor Michael, Jaron Mink, Jason Liu, Sneha Gaur, Wajih Ul Hassan, and Adam Bates. 2020. On the Forensic Validity of Approximated Audit Logs. In *Proceedings of the 36th Annual Computer Security Applications Conference (Austin, USA) (ACSAC '20)*. Association for Computing Machinery, New York, NY, USA, 189–202. doi:10.1145/3427228.3427272
- [26] Jessica Moosmann, Irind Pekaric, and Giovanni Apruzzese. 2026. Can SOC Operators Explain their Decisions while Triaging Alarms? A Real-World Study. *arXiv preprint arXiv:2604.22001* (2026).
- [27] Francis Sahngun Nahm. 2016. Nonparametric statistical tests for the continuous data: the basic concept and the practical use. *Korean journal of anesthesiology* 69, 1 (2016), 8–14.
- [28] Thomas Pasquier, Xueyuan Han, Mark Goldstein, Thomas Moyer, David Eysers, Margo Seltzer, and Jean Bacon. 2017. Practical Whole-System Provenance Capture. In *Proceedings of the 2017 Symposium on Cloud Computing (SoCC '17)*. ACM, 405–418. doi:10.1145/3127479.3129249
- [29] Thomas Pasquier, Xueyuan Han, Thomas Moyer, Adam Bates, Olivier Hermant, David Eysers, Jean Bacon, and Margo Seltzer. 2018. Runtime Analysis of Whole-System Provenance. In *Proceedings of the 2018 ACM SIGSAC Conference on Computer and Communications Security (CCS '18)*. ACM, 1601–1616. doi:10.1145/3243734.3243776
- [30] Nanda Rani, Bikash Saha, and Sandeep Kumar Shukla. 2025. A comprehensive survey of automated Advanced Persistent Threat attribution: Taxonomy, methods, challenges and open research problems. *Journal of Information Security and Applications* 92 (2025), 104076. doi:10.1016/j.jisa.2025.104076
- [31] Azadeh Tabiban, Heyang Zhao, Yosr Jarraya, Mekan Pourzandi, Mengyuan Zhang, and Lingyu Wang. 2022. ProvTalk: Towards Interpretable Multi-level Provenance Analysis in Networking Functions Virtualization (NFV). In *NDSS*.
- [32] Saimon Amanuel Tsegai, Xinyu Yang, Haoyuan Liu, and Peng Gao. 2025. Enabling Efficient Attack Investigation via Human-in-the-Loop Security Analysis. *Proceedings of the VLDB Endowment* 18, 11 (2025), 3771–3783. doi:10.14778/3749646.3749653
- [33] Wajih Ul Hassan, Shengjian Guo, Ding Li, Zhengzhang Chen, Kangkook Jee, Zhichun Li, and Adam Bates. 2019. NoDoze: Combatting Threat Alert Fatigue with Automated Provenance Triage. In *26th Annual Network and Distributed System Security Symposium (NDSS 2019)*. The Internet Society. doi:10.14722/ndss.2019.23349
- [34] Lingzhi Wang, Xiangmin Shen, Weijian Li, Zhenyuan Li, R Sekar, Han Liu, and Yan Chen. 2024. Incorporating gradients to rules: Towards lightweight, adaptive provenance-based intrusion detection. *arXiv preprint arXiv:2404.14720* (2024).
- [35] Frank Wilcoxon. 1945. Individual comparisons by ranking methods. *Biometrics bulletin* 1, 6 (1945), 80–83.
- [36] Yue Xiao, Ling Jiang, Sen Nie, Ding Li, Shi Wu, Ke Xu, and Qi Li. 2026. How Far Should We Need to Go : Evaluate Provenance-based Intrusion Detection Systems in Industrial Scenarios. arXiv:2603.22982 [cs.CR] <https://arxiv.org/abs/2603.22982>
- [37] Zhiqiang Xu, Pengcheng Fang, Changlin Liu, Xusheng Xiao, Yu Wen, and Dan Meng. 2022. DepComm: Graph Summarization on System Audit Logs for Attack Investigation. In *2022 IEEE Symposium on Security and Privacy (SP)*. IEEE, 540–557. doi:10.1109/SP46214.2022.9833632
- [38] Jun Zeng, Zheng Leong Chua, Yinfang Chen, Kaihang Ji, Zhenkai Liang, and Jian Mao. 2021. WATSON: Abstracting Behaviors from Audit Logs via Aggregation of Contextual Semantics. In *NDSS*.
- [39] Michael Zipperle, Florian Gottwalt, Elizabeth Chang, and Tharam Dillon. 2022. Provenance-based intrusion detection systems: A survey. *Comput. Surveys* 55, 7 (2022), 1–36.

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